

# Jyot Antani, PhD

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**Expertise:** Biophysics, Microbiology, Virology, Host-pathogen Interactions, Microscopy

## Experience & Education

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| Jun 2021-<br>Now      | Postdoctoral Research Associate, Yale University (US)<br><a href="#">Paul Turner Lab</a> , Department of Ecology and Evolutionary Biology<br><a href="#">Thierry Emonet Lab</a> , Molecular, Cellular, and Developmental Biology<br>Research: Viral target search on host surfaces - Dynamics, Evolution, and Therapeutic Application |
| Aug 2016-<br>May 2021 | PhD, <a href="#">Pushkar Lele Lab</a> , Department of Chemical Engineering, Texas A&M University (US)<br>Dissertation title: Sensory Functions of the Bacterial Flagellar Motor                                                                                                                                                       |
| Jul 2012-<br>May 2016 | Bachelor of Technology with Honours in Chemical Engineering<br>Indian Institute of Technology Bombay (IIT Bombay), Mumbai (India)                                                                                                                                                                                                     |

## Published Work

[Google Scholar Link](#)

- [Antani JD](#), Gupta R, Lee AH, Rhee KY, Manson MD, Lele PP (2021) Mechanosensitive recruitment of stator units promotes binding of the response regulator CheY-P to the flagellar motor. *Nature Communications*. doi: [10.1038/s41467-021-25774-2](https://doi.org/10.1038/s41467-021-25774-2)
- [Antani JD](#), Sumali AX, Lele TP, Lele PP (2021) Asymmetric random walks reveal that the chemotaxis network modulates flagellar rotational bias in *Helicobacter pylori*. *eLife*. doi:[10.7554/eLife.63936](https://doi.org/10.7554/eLife.63936)
- Katiyar A\*, [Antani JD](#)\*, McKee BP, Gupta R, Lele PP, Lele TP (2021) A method for direct imaging of x-z cross-sections of fluorescent samples. *Journal of Microscopy*. doi:[10.1111/jmi.12965](https://doi.org/10.1111/jmi.12965) (\*equal contribution)
- Wong GCL, [Antani JD](#), et al. (2021) Roadmap on emerging concepts in the physical biology of bacterial biofilms: from surface sensing to community formation. *Physical Biology*. doi:[10.1088/1478-3975/abdc0e](https://doi.org/10.1088/1478-3975/abdc0e)
- Ford KM, [Antani JD](#), Nagarajan A, Johnson MM, Lele PP (2018) Switching and torque generation in swarming cells of *E. coli*. *Frontiers in Microbiology*, 9, 2197. doi:[10.3389/fmicb.2018.02197](https://doi.org/10.3389/fmicb.2018.02197)
- [Antani JD](#), Gupta R, Lele PP. The chemotaxis conundrum for a polar-flagellated bacterium. *In preparation*.
- Katiyar A, Zhang J, [Antani JD](#), Yu Y, Lele PP, Reinhart-King CA, Sniadecki NJ, Roux KJ, Dickinson RB, Lele TP (2022) Lamin A/C mediated invaginations in the nuclear surface allow the nucleus to pass unimpeded through a dense array of fiber-like obstacles. *Advanced Science*, Accepted. Preprint doi: [10.1101/2022.03.10.483838](https://doi.org/10.1101/2022.03.10.483838)

## Peer Review (Independent)

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|----------|-------------------------------------------|
| Dec 2021 | <i>European Biophysics Journal (EBOJ)</i> |
| May 2022 | <i>Physical Biology</i>                   |

## Selected Conference Presentations

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|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <a href="#">Antani JD</a> , Emonet T, Turner PE. Dynamic target search by bacterial viruses on host surfaces |                                                                                                 |
| Jun 2022                                                                                                     | <i>Microbe</i> - Annual meeting of the American Society for Microbiology, Washington, D.C. (US) |
| May 2022                                                                                                     | Yale Postdoctoral Association Symposium, Yale University, New Haven (US)                        |
| <a href="#">Antani JD</a> , Lele PP. Modulation of chemotactic signaling in bacteria by mechanical forces.   |                                                                                                 |
| Mar 2019                                                                                                     | ChEGSA Annual Symposium, Texas A&M University, College Station (US)                             |
| Oct 2018                                                                                                     | American Institute of Chemical Engineers (AIChE) Annual Meeting, Pittsburgh (US)                |

Antani JD, Lele PP. Asymmetric Swimming and Chemotaxis in *Helicobacter pylori*.

Jan 2021 | Bacterial Locomotion And Signal Transduction (BLAST) XVI Meeting, Zoom (online)  
Feb 2020 | Biophysical Society (BPS) Annual Meeting, San Diego (US).  
Jan 2019 | Bacterial Locomotion And Signal Transduction (BLAST) XV Meeting, New Orleans (US)

Antani JD, Venkatesh KV. Analysis of Glucose Homeostasis in Humans.

Nov 2015 | Futuristic Approach for Alternatives to Animal Testing [CEFIPRA](#) seminar, IIT Bombay, India

## Honors & Awards

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|------|-----------------------------------------------------------------------------------------------|
| 2020 | CIRTL Associate Certificate, Center for the Integration of Research, Teaching, and Learning   |
| 2015 | Best Poster Award, <i>AMAT</i> seminar, IIT Bombay                                            |
| 2015 | Undergraduate Research Award URA01, IIT Bombay                                                |
| 2012 | All India Rank 1427 in Joint Entrance Examination for engineering, among 479,651 participants |
| 2008 | National Talent Search Scholarship, NCERT, India                                              |

## Technical Skills

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|-----------------------------------------|---------------------------|---------------------------------------|
| Perfusion & Thermal Control Experiments | Prototyping & 3D-printing | Microscopy: Dark-field, TIRF, Optical |
| Bacteria+Phage Culture, Cloning, PCR    | Microfluidics Experiments | Tweezers, Image Analysis              |

## Computer Proficiency / Computational Skills

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|------------------------------|----------------------------------------------------------------------------------|
| Image Processing & Analysis: | MATLAB, Python, ImageJ – {Highlight: particle-tracking algorithms}               |
| Genetic Analysis:            | Molecular Biology {SnapGene, Benchling}, Whole Genome Analysis { <i>breseq</i> } |
| Design & Development:        | SolidWorks, AutoCAD, LabVIEW,                                                    |
| Simulation Packages:         | ANSYS Fluent, Aspen Plus, OpenFOAM, COMSOL, CHARMM, VMD                          |
| Programming:                 | MATLAB, Python, C, C++, Fortran, HTML                                            |

## Current and Doctoral Research

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|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| May 2021-<br>Now      | <p>Postdoctoral Research: <b>Host-virus Interaction Biophysics in Bacteria</b><br/><i>Evolution of target search by bacteriophages on host bacterial surfaces</i></p> <ul style="list-style-type: none"><li>Established a protocol for labeling phages (the viruses of bacteria) with a non-specific fluorescent dye</li><li>Visualizing labeled phage (bacterial virus) particles attaching to host cells, through epifluorescence and HILO microscopy</li><li>Set up MATLAB-based statistical analysis of single phage trajectories to quantify attachment dynamics</li><li>Employing Bayesian inference to estimate the diffusive states of phage particles from single trajectories</li><li>Comparing the attachment dynamics of ancestral and coevolved phage-host pairs</li></ul> <p><i>Evolution of resistance in E. coli against a flagellotropic phage</i></p> <ul style="list-style-type: none"><li>Evolved mutants resistant against phage <math>\chi</math>, a virus that uses rotating bacterial flagella as receptors</li><li>Trained an undergraduate to carry out the experimental evolution protocol, mentored him in writing a successful fellowship application about this project</li><li>Characterizing the genotype and phenotype of the evolved mutants to determine the evolutionary pathway(s)</li></ul> <p><i>Development of phage therapy against gastric cancer agent Helicobacter pylori</i></p> <ul style="list-style-type: none"><li>Devising a method to isolate novel phages that infect <i>H. pylori</i>, a human pathogen</li><li>Wrote+writing several fellowship and grant applications to fund the original research projects</li></ul> |
| Aug 2016-<br>May 2021 | <p>Doctoral Thesis: <b>Sensory Functions of the Bacterial Flagellar Motor</b><br/><i>Molecular mechanisms of host substrate sensing by bacterial flagellar motors</i></p> <ul style="list-style-type: none"><li>Determined the effect of mechanical forces on chemotactic output of the bacterial flagellar motor, with phase microscopy, optical tweezers, and TIRF imaging</li><li>Developed a biophysical model for the regulation of allosteric interactions by mechanical force</li></ul> <p><i>Motility, chemotaxis, and random spread of Helicobacter pylori, the carcinogenic pathogen</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

- Devised a microscopy-based technique to experimentally determine the chemotactic activity of *H. pylori*
- Studied the biophysical principles governing chemotactic migration by this bacterial species which employs run-reversal motility – different from the traditionally studied *E. coli*

#### Design and development of experimental and analytical framework

- Designed 3D-printed imaging chambers for specific requirements in microscopy experiments, e.g., perfusion, imaging side-views
- Wrote MATLAB-based image analysis codes for tracking motile bacteria, mammalian cell-nuclei, and fluorophores

## Additional Research Experience

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|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jul 2015-<br>Feb 2016     | <b>Machine Learning Classification of Glucose Homeostasis in Diabetic and Healthy Individuals</b><br><b>Venkatesh Lab, Chemical Engineering, IIT Bombay</b> <ul style="list-style-type: none"> <li>• Formulated a Hill function-based empirical biochemical model of body glucose concentration following a fixed food-intake, obtained least-squares fits to data acquired from 150+ individuals</li> <li>• Constructed a classification tree based on fit-parameters using machine learning in MATLAB; established a statistical grouping among Diabetic and Healthy individuals</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| May-Jul<br>2015           | <b>A Novel Protocol for Coating Lipid Bilayers onto Silver Nanocubes</b><br><b>Hung-Jen Wu Lab, Texas A&amp;M University</b> <ul style="list-style-type: none"> <li>• Established a tip-sonication protocol to make small unilamellar vesicles of lipids</li> <li>• Coated the unilamellar vesicles as supported lipid bilayers onto Silver nanocubes</li> <li>• Employed a Localized Surface Plasmon Resonance-based nanocube sensor to monitor protein binding on lipid bilayers</li> <li>• Analyzed binding kinetics of GM1 ganglioside with Cholera Toxin Subunit B</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| May 2014-<br>Apr 2015     | <b>Simulation-Design of a Chemical Vapor Deposition Reactor</b><br><b>Mahesh Tirumkudulu Lab, IIT Bombay</b> <ul style="list-style-type: none"> <li>• Simulated fluid flow around a disk, successfully obtained the pressure coefficients observed experimentally in literature.</li> <li>• Proposed an unconventional design for uniform coating of parylene polymer onto silicon micro-wafers in a micro-scale Chemical Vapor Deposition reactor.</li> <li>• Modeled Parylene flow using ANSYS Fluent, enhanced the proposed design.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Jun 2013                  | <b>Process Improvement and Technology in Jaggery-Making</b><br><b>Sanjay Mahajani Lab, IIT Bombay</b> <ul style="list-style-type: none"> <li>• Examined units under development for continuous production as well as recuperation in manufacturing Jaggery, the traditional non-centrifugal cane sugar. Analyzed Jaggery samples for acidity, moisture content, water activity, density, sugar content etc; observed the XRD and ESEM techniques.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Key<br>Course<br>Projects | <b>Determining Diffusion Coefficient of Microparticles using Particle-Tracking Algorithms</b><br><i>Course: Colloidal Applications and Biomolecular Interactions in Life Science (Spring 2018)</i> <ul style="list-style-type: none"> <li>• Employed particle-tracking algorithms in MATLAB to track the Brownian diffusion of submicron-sized particles recorded using phase-contrast microscopy</li> <li>• Determined the diffusion coefficient of particles by calculating the average mean-squared displacement of an ensemble</li> </ul><br><b>Proposal: Evolution of Bacterial Flagellum</b> <i>Course: Evolutionary Bioinformatics (Spring 2017)</i> <ul style="list-style-type: none"> <li>• Proposed a study involving the comparison of flagellar genome across multiple bacterial species which has potential to aid in obstruction of pathogenic motility</li> <li>• Carried out BLAST alignment analysis for nucleotide and protein sequences of flagellin (flagellar filament) gene as preliminary results</li> </ul><br><b>Review: How Deep a Cell Can Sense</b> <i>Course: Cell Mechanics (Spring 2016)</i> <ul style="list-style-type: none"> <li>• Carried out a literature survey and reported theoretical, in-vitro, and in-silico studies on quantified investigations about the range of mechanical signaling</li> </ul><br><b>On-line Estimation of Concentrations in a Bio-Reactor</b> <i>Course: State Estimation (Fall 2015)</i> <ul style="list-style-type: none"> <li>• Generated true states from a model based on substrate uptake, phosphorus repression and effect of seed culture on glyco-peptide antibiotic production</li> <li>• Implemented the Simple, Extended, and Unscented versions of the Kalman Filter to estimate the concentrations</li> </ul> |

## Teaching & Mentoring Experience

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Fall 2019 | A hands-on workshop for image processing and single particle tracking in MATLAB  
*Pushkar Lele Lab, Texas A&M University*

### Undergraduate students mentored

Oct 2021-  
Now | [Austin Theroux](#), Ecology & Evolutionary Biology, Yale University  
Studying the evolution of motile bacteria against a bacteriophage that attaches through the flagellum  
Presenting a poster about this work at *Microbe* ASM meeting at Washington DC in June, 2022

Dec 2019-  
Jan 2021 | [Anita Sumali](#), Biomedical Engineering, Texas A&M University  
MATLAB-based quantitative analysis of the run-reversal motility of *H. pylori*; co-author on [Antani et al., eLife, 2021](#)

Apr 2017-  
Apr 2018 | AggieE\_Challenge – Undergraduate Mentoring, Texas A&M University  
Graduate-mentor to a multidisciplinary team of eight undergraduate students employing 3D-printing technology to design scaled-up bacterial models, in order to study the hydrodynamic effects of cell morphology on motility

### Teaching Assistant at Texas A&M Chemical Engineering

Spring 2019 | Heat Transfer Operations  
Spring 2017 | Bioprocess Engineering  
Fall 2016 | Bioprocess Engineering

## Teaching, Mentoring, & Management Training

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Sep 2021-  
Dec 2021 | Auburn University's Preparing Future Faculty Workshop  
Interactive workshops consisting panel discussions from faculty and university administrators

Sep 2021 | [Fundamentals of Inclusive Teaching](#)  
[Poorvu Center: Graduate and Postdoctoral Teaching Development, Yale University](#)  
A 4-part workshop on creating a respectful, supportive, equitable, and transparent classroom environment

Aug 2018-  
Dec 2019 | [Academy for Future Faculty \(AFF\)](#), Center for Teaching Excellence, Texas A&M University  
*Faculty Mentor: Dr. Micah Green, Texas A&M University*  
A program designed for training graduate students as future faculty members, featuring weekly seminars and workshops, and anchored by faculty mentorship

Jul-  
Aug 2019 | An Introduction to Evidence-Based Undergraduate STEM Teaching  
*Multi-university Massive Open Online Course from the NSF-funded CIRTl network*  
Eight-week online course on [edX.org](#) catering introduction to effective teaching strategies and the research that supports them. Attended local learning community meetings for the course at Texas A&M University.

Oct 2019 | Improving Research Mentoring Relationships – Introduction to Effective Mentorship  
*Center for the Improvement of Mentored Experiences in Research (CIMER)*  
1-day workshop at Texas A&M University, conducted by CIMER from University of Wisconsin-Madison

## Leadership & Management Experience

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Jan 2022-  
Now | Communications Manager, [Yale Postdoctoral Association](#)

- Managing website profiles and social media for the YPA
- Linking outside organizations to the YPA community

Jun 2021 | [People Management Skills](#) – Course, Chartered Institute of Personnel and Development, UK

Oct 2018-  
Aug 2020 | [Secretary \(2018-19\) & Chair \(2019-20\), Chemical Engineering Graduate Student Advisory Council](#)  
*Artie McFerrin Department of Chemical Engineering, Texas A&M University*

- Designed the council by-laws as a founding member
- Set up channels of communication through surveys and anonymous feedback mechanisms, between grad students and administration in the department
- Addressed grad student welfare issues including salary, benefits, mental well-being, work-life balance, student-advisor relationships, facilities, career events, and social events

Apr 2014-  
Mar 2015

Competitions Manager, AZeotropy 2015

*IIT Bombay's Intercollegiate Chemical Engineering Symposium*

- Pioneered a new model-making competition *ChemETimer* based on time-precision
- Designed a presentation-based competition *Swaasthya* on engineering-applications to healthcare
- Designed *Cipher*, the online cryptic hunt which received a participation of 400+
- Actualized unconventional chemical engineering quizzing for *Chem-O-Philia*, national level quiz

## Science Outreach Volunteering Experience

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Feb 2022-  
Now

[Associate Writer & Editor, LabLeaks](#)

A biweekly mailing list summarizing recent noteworthy scientific articles to the general audience.

Feb 2022

[Exploring Science for Middle School Students](#)

Presented research topic to middle school children in the New Haven County, Connecticut

Jul 2020

[Coronasurveys: Monitoring the Incidence of COVID-19 via Open Surveys](#)

Volunteered services as the primary translator of the survey to the Gujarati language.